

Theorem (5.20)

If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

FIGURE 5.19

Nonintegrable discontinuous function

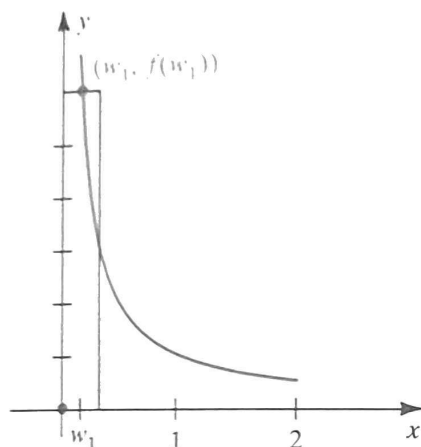
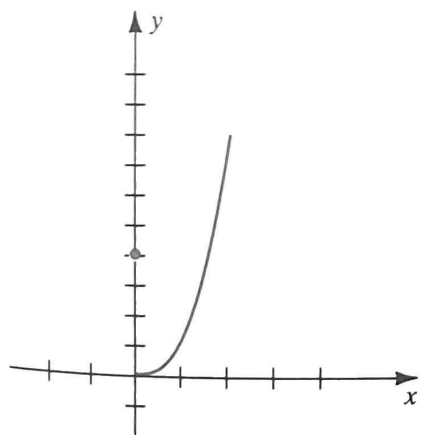


FIGURE 5.20

Integrable discontinuous function



A proof of Theorem (5.20) may be found in texts on advanced calculus.

Definite integrals of discontinuous functions may or may not exist, depending on the types of discontinuities. In particular, *functions that have infinite discontinuities on a closed interval are not integrable on that interval.* To illustrate, consider the piecewise-defined function f with domain $[0, 2]$ such that

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ \frac{1}{x} & \text{if } 0 < x \leq 2 \end{cases}$$

The graph of f is sketched in Figure 5.19. Note that $\lim_{x \rightarrow 0^+} f(x) = \infty$. If M is any (large) positive number, then in the first subinterval $[x_0, x_1]$ of any partition P of $[a, b]$, we can find a number w_1 such that $f(w_1) > M/\Delta x_1$, or, equivalently, $f(w_1) \Delta x_1 > M$. It follows that there are Riemann sums $\sum_k f(w_k) \Delta x_k$ that are arbitrarily large, and hence the limit in Definition (5.16) cannot exist. Thus, f is not integrable. A similar argument can be given for any function that has an infinite discontinuity in $[a, b]$. Consequently, *if a function f is integrable on $[a, b]$, then it is bounded on $[a, b]$; that is, there is a real number M such that $|f(x)| \leq M$ for every x in $[a, b]$.*

As an illustration of a discontinuous function that is integrable, consider the piecewise-defined function f with domain $[0, 2]$ such that

$$f(x) = \begin{cases} 4 & \text{if } x = 0 \\ x^3 & \text{if } 0 < x \leq 2 \end{cases}$$

The graph of f is sketched in Figure 5.20. Note that f has a jump discontinuity at $x = 0$. From Example 7 of Section 5.3, the area under the graph of $y = x^3$ from 0 to 2 is $2^4/4 = 4$. Thus, by Theorem (5.19), $\int_0^2 x^3 dx = 4$. We can also show that $\int_0^2 f(x) dx = 4$. Hence f is integrable.

We have shown that a function that is discontinuous on a closed interval may or may not be integrable. However, by Theorem (5.20), functions that are *continuous* on a closed interval are *always* integrable.

EXERCISES 5.4

Exer. 1–4: The given numbers determine a partition P of an interval. (a) Find the length of each subinterval of P . (b) Find the norm $\|P\|$ of the partition.

- 1 $\{0, 1.1, 2.6, 3.7, 4.1, 5\}$
- 2 $\{2, 3, 3.7, 4, 5.2, 6\}$
- 3 $\{-3, -2.7, -1, 0.4, 0.9, 1\}$
- 4 $\{1, 1.6, 2, 3.5, 4\}$

Exer. 5–6: Let P be the partition of $[1, 5]$ determined by $\{1, 3, 4, 5\}$. Find a Riemann sum R_P for the given function f by choosing, in each subinterval of P , (a) the

right-hand endpoint, (b) the left-hand endpoint, and (c) the midpoint.

5 $f(x) = 2x + 3$

6 $f(x) = 3 - 4x$

7 If $f(x) = 8 - \frac{1}{2}x^2$ and P is the regular partition of $[0, 6]$ into six subintervals, find a Riemann sum R_P of f by choosing the midpoint of each subinterval.

8 If $f(x) = 8 - \frac{1}{2}x^2$ and P is the partition of $[0, 6]$ determined by $\{0, 1.5, 3, 4.5, 6\}$, find a Riemann sum R_P of f by choosing the numbers 1, 2, 4, and 5 in the subintervals of P .

- 9 If $f(x) = x^3$ and P is the partition of $[-2, 4]$ determined by $\{-2, 0, 1, 3, 4\}$, find a Riemann sum R_P of f by choosing the numbers $-1, 1, 2$, and 4 in the subintervals of P .
- 10 If $f(x) = \sqrt{x}$ and P is the partition of $[1, 16]$ determined by $\{1, 3, 5, 7, 9, 16\}$, find a Riemann sum R_P of f by choosing the numbers $1, 4, 5, 9$, and 9 in the subintervals of P .
- [c] 11 If $f(x) = x^2 \sqrt{\cos x}$ and P is the regular partition of $[0, 1]$ into ten subintervals, find a Riemann sum R_P of f by choosing the midpoint of each subinterval of P .
- [c] 12 If $f(x) = \sin(\cos x)$ and P is the partition of $[-1, 1]$ determined by $\{-1, -0.65, -0.23, 0.51, 0.85, 1\}$, find a Riemann sum R_P of f by choosing the numbers $-0.75, -0.5, 0, 0.6, 0.9$ in the subintervals of P .

Exer. 13–16: Use Definition (5.16) to express each limit as a definite integral on the given interval $[a, b]$.

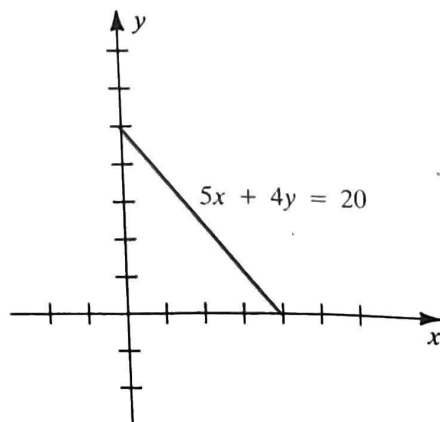
- 13 $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (3w_k^2 - 2w_k + 5) \Delta x_k; \quad [-1, 2]$
- 14 $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \pi(w_k^2 - 4) \Delta x_k; \quad [2, 3]$
- 15 $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n 2\pi w_k(1 + w_k^3) \Delta x_k; \quad [0, 4]$
- 16 $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (\sqrt[3]{w_k} + 4w_k) \Delta x_k; \quad [-4, -3]$

Exer. 17–22: Given $\int_1^4 \sqrt{x} \, dx = \frac{14}{3}$, evaluate the integral.

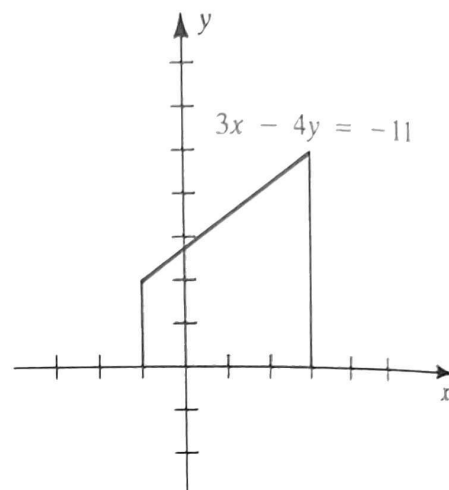
- 17 $\int_4^1 \sqrt{x} \, dx$ 18 $\int_1^4 \sqrt{s} \, ds$
- 19 $\int_1^4 \sqrt{t} \, dt$ 20 $\int_1^4 \sqrt{x} \, dx + \int_4^1 \sqrt{x} \, dx$
- 21 $\int_4^4 \sqrt{x} \, dx + \int_4^1 \sqrt{x} \, dx$ 22 $\int_4^4 \sqrt{x} \, dx$

Exer. 23–26: Express the area of the region in the figure as a definite integral.

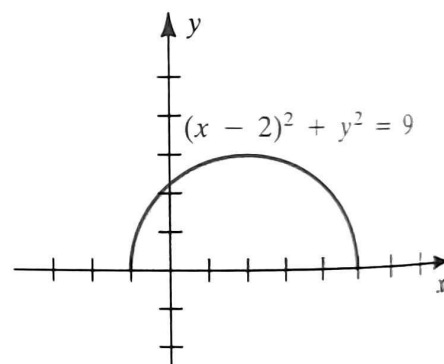
23



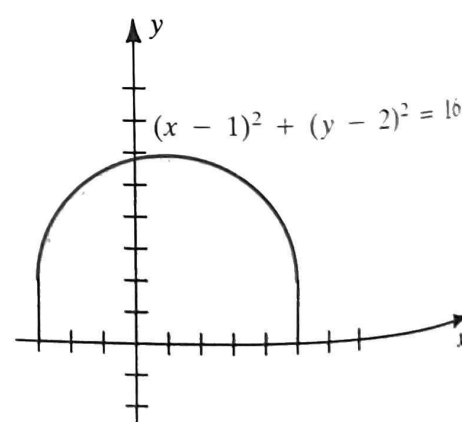
24



25



26



Exer. 27–36: Evaluate the definite integral by regarding it as the area under the graph of a function.

- 27 $\int_{-1}^5 6 \, dx$ 28 $\int_{-2}^3 4 \, dx$
- 29 $\int_{-3}^2 (2x + 6) \, dx$ 30 $\int_{-1}^2 (7 - 3x) \, dx$
- 31 $\int_0^3 |x - 1| \, dx$ 32 $\int_{-1}^4 |x| \, dx$
- 33 $\int_0^3 \sqrt{9 - x^2} \, dx$
- 34 $\int_0^a \sqrt{a^2 - x^2} \, dx, \quad a > 0$
- 35 $\int_{-2}^2 (3 + \sqrt{4 - x^2}) \, dx$
- 36 $\int_{-2}^2 (3 - \sqrt{4 - x^2}) \, dx$